

THE SCOTS COLLEGE



YEAR 12

TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

EXTENSION TWO MATHEMATICS

AUGUST 2002

TIME ALLOWED: 3 HOURS [plus 5 minutes reading time]

OUTCOMES ASSESSED:

- E3 - Uses the relationship between algebraic and geometric representations of complex numbers and of conic sections.
- E4 - Uses efficient techniques for algebraic manipulation required in dealing with conic sections and polynomials.
- E6 - Combines the ideas of algebra and calculus to determine the important features of the graphs of a wide variety of functions.
- E7 - Uses techniques in slicing and volumes. Applies further techniques of integration to problems.
- E8 - Applies further techniques of integration.
- E5 - Uses ideas and techniques from calculus to solve problems in mechanics involving resolution of forces, resisted motion and circular motion.

QUESTION 1 [15 MARKS]

- a. Given the complex number $z = 7 - 3i$. Find
- (i) $|z|$
 - (ii) $\bar{z}$
 - (iii) $|z - \bar{z}|$
 - (iv) $\arg(z - \bar{z})$
- b. Express $z = \frac{\sqrt{2}}{1-i}$ in modulus argument form and hence express z^5 in the form $x + iy$.
- c. K, L, M, N are vertices of a square, in anti-clockwise order. Given that K and M represent the numbers $2 + i$ and $2 + 3i$ respectively, find the coordinates of:
- (i) L and N.
 - (ii) M, if the square is rotated clockwise through an angle of 90° about the origin.
- d. In the Argand Plane, sketch the following:
- (i) $|z - 12 + 3i| = 5$
 - (ii) $|z^2 - \bar{z}^2| \geq 4$
 - (iii) $\arg \frac{z-1+i}{z+1-i} = 0$

START A NEW BOOKLET

QUESTION 2 [15 MARKS]

- a. The polynomial function $p(x) = x^4 - 4x^3 - 3x^2 + 50x - 52$ has a zero at $x = 3 - 2i$. Factorise $P(x)$ over the field of:
- (i) rationals
 - (ii) reals
 - (iii) complex numbers

- b. The equation $x^4 + 4x^3 - 3x^2 - 4x - 2 = 0$ has roots $\alpha, \beta, \gamma, \delta$. Find the equation with roots:

$$\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}, \frac{1}{\delta}$$

- c. The equation $2x^3 - 9x^2 + 7 = 0$ has roots α, β, γ . Find the equation with root $\alpha^3, \beta^3, \gamma^3$.

- d. Solve $x^5 + 2x^4 - 2x^3 - 8x^2 - 7x - 2 = 0$ if it has a root of multiplicity 4.

START A NEW BOOKLET

QUESTION 3 [15 MARKS]

- a. Find the coordinates of the two foci on the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$
- b. P and Q are variable points on the rectangular hyperbola $xy = c^2$.
- The tangent at Q passes through the foot of the ordinate of P. If P and Q have parameters p and q , show that $p = 2q$.
 - Hence prove that the locus of the midpoint of PQ is a rectangular hyperbola and find its equation.
- c. The hyperbola H has equation $xy = 16$.
- Sketch this hyperbola and indicate on your diagram the positions and coordinates of all points at which the curve intersects the axes of symmetry.
 - $P\left(4p, \frac{4}{p}\right)$, where $p > 0$, and $Q\left(4q, \frac{4}{q}\right)$ where $q > 0$ are two distinct arbitrary points on H. Find the equation of chord PQ.
 - Prove that the equation of the tangent at P is $x + p^2y = 8p$.
 - The tangents at P and Q intersect at T. Find the coordinates of T.

QUESTION 4 [15 MARKS]

- a. Find:

(i) $\int \tan^{-1} x \, dx$

(ii) $\int \frac{dx}{\sin x \cos x}$

(iii) $\int \frac{\sec^2 x \, dx}{\tan^2 x - 3 \tan x + 2}$

- b. Leaving your answer in exact form, evaluate:

(i) $\int_0^1 \frac{dx}{x^2 + 8x + 4}$

(ii) $\int_0^1 x^2 e^{-x} \, dx$

START A NEW BOOKLET

QUESTION 5 [15 MARKS]

Sketch the following curves on separate axes, showing all relevant points.

- a. (i) $y = \sin x$ and hence also sketch;
 (ii) $y^2 = \sin x$ for $-2\pi \leq x \leq 2\pi$.

- b. (i) $y = x^3 - 4x$ and hence also sketch;
 (ii) $y = |x^3| - 4|x|$ for Domain $-3 \leq x \leq 3$.

- c. Sketch the graph of $y = \frac{x^2 - 6x + 8}{x^2 - x - 6}$ clearly indicating all relevant points.

QUESTION 6 [15 MARKS]

- a. Determine the real values for k for which the equation

$$\frac{x^2}{19-k} + \frac{y^2}{7-k} = 1$$

defines respectively an ellipse and a hyperbola. Sketch the curve corresponding to the value $k = 3$.

- b. Evaluate the definite integral

$$\int_{-1}^1 \frac{4+x^2}{4-x^2} dx$$

- c. Sketch the region (R) which is completely bounded by the curves $y = \sin 2x$ and $y = \frac{1}{2}$ in the domain, $0 \leq x \leq \frac{\pi}{2}$. Find the volume generated when R is rotated about the:

(i) x -axis

(ii) line $y = \frac{1}{2}$

START A NEW BOOKLET

QUESTION 7 [15 MARKS]

- a. The base of a solid is a circular region of radius a units. Find the volume if every cross-section of a plane perpendicular to a certain diameter is a square with one side lying in the base.
- b. Find, by the method of cylindrical shells, the volume of the solid generated when the region bounded by the curve $y = x^2 + 1$, the line $x = 2$ and the coordinate axes is rotated about the line $x = 3$.

QUESTION 8 [15 MARKS]

- a. A parachutist of mass m falls to ground from an aircraft. Given that the air resistance per unit mass is proportional to the square of the speed V :

(i) Draw a diagram showing clearly the forces acting on the parachutist during his free fall.

(ii) Deduce that $\frac{d}{dx}(v^2) = 2g - 2kv^2$

(iii) Show that $v^2 = \frac{g}{k} - Ae^{-2kx}$ satisfies the differential equation in part (ii) above and show that $A = \frac{g}{k}$.

(iv) Sketch the graph of v^2 against x and find an expression for the terminal speed of the parachutist during his free fall.

- b. A particle of mass 10kg is found to experience a resistance, in Newtons, one tenth of the square of its velocity in m/sec, when it moves through the air. The particle is projected vertically upwards from 0 with a velocity of u m/sec, and the point A, vertically above 0, is the highest point reached by the particle before it starts to fall to the ground again. Assuming the value of g is 10 m/sec².

(i) Find the time taken for the particle to reach A from 0.

(ii) Show that the height OA is $50 \ln(1 + 10^{-3}u^2)$.

(iii) Show that the particles velocity W m/sec when it reaches 0 again is given by

$$W^2 = \frac{u^2}{1 + 10^{-3}u^2}$$

Standard Integrals

$$\int x^n dx = \frac{1}{n+1} x^{n+1} + C, \quad n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x + C, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax} + C, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax + C, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax + C, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax + C, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax + C, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + C, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}) + C, \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2}) + C, \quad a \neq 0$$

NOTE: $\ln x \equiv \log_e x, \quad x > 0$

Soln.
 2002
 Final
 Soln.

[Question 1]

a) Given $z = 7 - 3i$

i) $|z| = \sqrt{49 + 9}$
 $= \sqrt{58}$ 1

ii) $\bar{z} = 7 + 3i$ 1

iii) $|z - \bar{z}|$
 $= |-6i|$
 $= 6$ 1

iv) $\arg(z - \bar{z})$
 $= \arg(-6i)$
 $= -\frac{\pi}{2}$ 1

b) Express $z = \frac{\sqrt{2}}{1-i}$ in

Mod
 arg form

$z = \frac{\sqrt{2}}{1-i} \times \frac{(1+i)}{(1+i)}$

$= \frac{\sqrt{2}(1+i)}{2}$

$= \frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}}$ 2

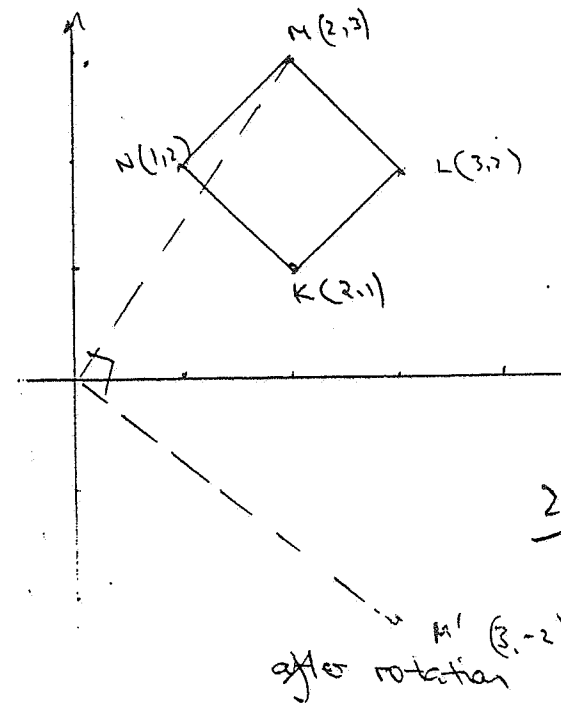
$= \cos \frac{\pi}{4} + i \sin \frac{\pi}{4}$

$\Rightarrow z^5 = \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)^5$

$= \cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4}$

$= -\cos \frac{\pi}{4} - i \sin \frac{\pi}{4}$

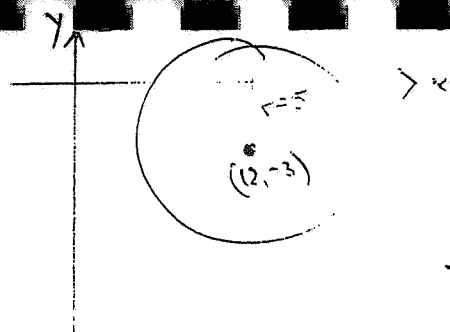
$= -\frac{1}{\sqrt{2}} - \frac{i}{\sqrt{2}}$ 2



$$i) |z - 12 + 3i| = 5$$

Circle centre (12, -3)

radius 5



$$ii) |z^2 - \bar{z}^2| \geq 4$$

$$|(z + \bar{z})(z - \bar{z})| \geq 4$$

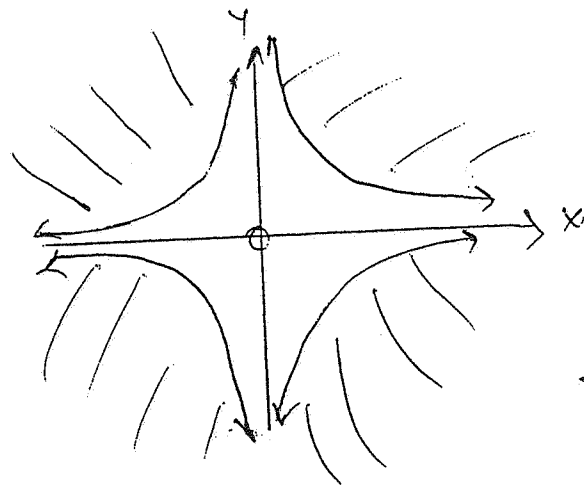
$$|(2x)(2iy)| \geq 4$$

$$|xyi| \geq 1$$

$$|xy| \geq 1$$

consider $xy = 1$

and $xy = -1$



$$iii) \arg \frac{z-1+i}{z+1-i} = 0$$

$$\arg(z-1+i) - \arg(z+1-i) = 0$$

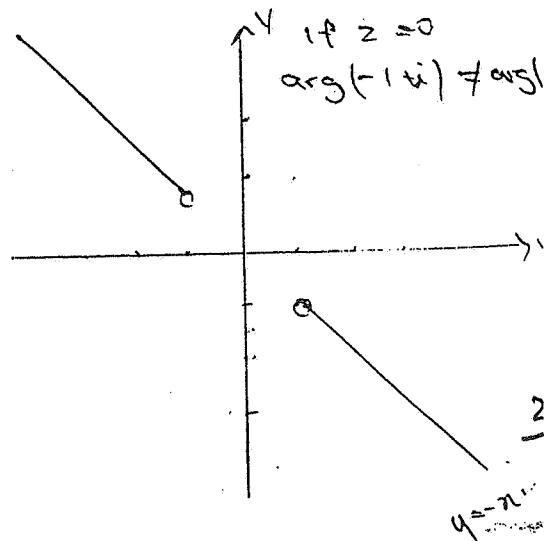
$$\tan^{-1} \frac{y+1}{x-1} = \tan^{-1} \frac{y-1}{x+1}$$

$$\therefore \frac{y+1}{x-1} = \frac{y-1}{x+1}$$

$$2x + 2y = 0$$

$$x + y = 0$$

$$y = -x$$



iii) For the particle on the return trip, let the origin and the direction be the same as above

$$ma = -mg - \frac{V^2}{10}$$

$$a = v \frac{dv}{dx} = -10 + \frac{V^2}{100}$$

$$\frac{v dv}{-10 + \frac{v^2}{100}} = \frac{100 v dv}{v^2 - 1000} = dx$$

$$\int_0^{-w} \frac{100 v dv}{v^2 - 1000} = \int_H^0 dx$$

$$-H = 50 \left[\ln(v^2 - 1000) \right]_0^{-w}$$

$$= 50 \ln \frac{w^2 - 1000}{-1000}$$

$$\therefore H = -50 \ln \left(1 - \frac{w^2}{1000} \right)$$

$$\therefore 50 \ln \left(1 + \frac{w^2}{1000} \right) = -50 \ln \left(1 - \frac{w^2}{1000} \right)$$

$$\Rightarrow \ln \left(1 + \frac{w^2}{1000} \right) = -\ln \left(1 - \frac{w^2}{1000} \right)$$

END SOLUTION

Question Two

a) $P(x) = x^4 - 4x^3 - 3x^2 + 50x - 52$

has a zero value at $x = 3 - 2i$

If $x = 3 - 2i$ is a factor then $x = 3 + 2i$

is also a factor.

i.e. $(x - 3 + 2i)(x - 3 - 2i)$ is a factor.

$\therefore x^2 - 6x + 13$ is a factor.

$\therefore$ i) $P(x) = (x^2 - 6x + 13)(x^2 + 2x - 4)$

over a rational field

ii) Consider from above $x^2 + 2x - 4$

$$x = \frac{-2 \pm \sqrt{4 - 4 \times (-4)}}{2}$$

$$= \frac{-2 \pm \sqrt{20}}{2}$$

$$= \frac{-2 \pm 2\sqrt{5}}{2}$$

$$x = -1 \pm \sqrt{5}$$

$$P(x) = (x^2 - 6x + 13)(x + 1 - \sqrt{5})(x + 1 + \sqrt{5})$$

over real field

iii) $(x - 3 + 2i)(x - 3 - 2i)(x + 1 - \sqrt{5})(x + 1 + \sqrt{5})$

over complex field.

$$\alpha, \beta, \gamma, \delta$$

New equation has roots

$$x = \frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}, \frac{1}{\delta}$$

$$\therefore \alpha = \frac{1}{x}$$

Therefore new eqnⁿ is

$$\Rightarrow \left(\frac{1}{x}\right)^4 + 4\left(\frac{1}{x}\right)^3 - 3\left(\frac{1}{x}\right)^2 - 4\left(\frac{1}{x}\right) - 2 = 0$$

$$\Rightarrow \frac{1}{x^4} + \frac{4}{x^3} - \frac{3}{x^2} - \frac{4}{x} - 2 = 0 \quad 3$$

$$\Rightarrow 1 + 4x - 3x^2 - 4x^3 - 2x^4 = 0$$

i) $2x^3 - 9x^2 + 7 = 0$ has roots α, β, γ

Find equation with roots $\alpha^3, \beta^3, \gamma^3$

New equation has roots

$$x = \alpha^3, \beta^3, \gamma^3$$

$$\therefore x^{1/3} = \alpha$$

$$10 + v^2 = 100$$

$$100 \int_0^6 \frac{dv}{1000 + v^2} = \int_0^t -dt$$

$$\therefore t = \sqrt{10} \tan^{-1} \frac{u\sqrt{10}}{100}$$

$$11) a = v \frac{dv}{dx} = -10 - \frac{v^2}{100}$$

$$\frac{v dv}{10 + v^2} = \frac{100v dv}{1000 + v^2} = -dx$$

$$100 \int_0^H \frac{v dv}{1000 + v^2} = \int_0^H -dx$$

$$-H = 50 \left[\ln(1000 + v^2) \right]_0^H$$

$$= 50 \ln \frac{1000}{1000 + u^2}$$

$$\therefore H = 50 \ln \frac{1000 + u^2}{1000}$$

$$= 50 \ln \left(1 + \frac{u^2}{1000} \right)$$

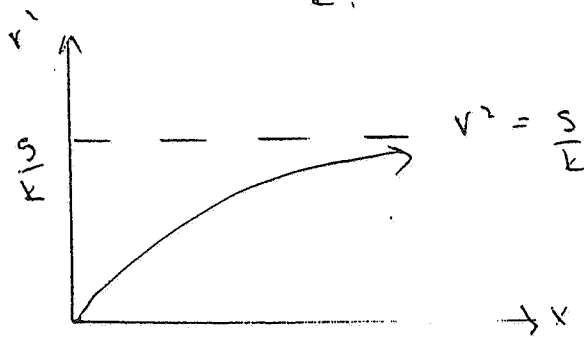
from $v = \frac{g}{k} - Ae^{-2kx}$ $A = \frac{g}{k}$

$$v^2 = \frac{g}{k} - \frac{g}{k} e^{-2kx}$$

$$v^2 = \frac{g}{k} (1 - e^{-2kx})$$

$$\lim_{x \rightarrow \infty} v^2 = \lim_{x \rightarrow \infty} \frac{g}{k} (1 - e^{-2kx})$$

$$v^2 = \frac{g}{k} \Rightarrow \text{terminal speed} = \sqrt{\frac{g}{k}}$$



86) 1) Let the upward direction be +ve and 0 be origin

$$ma = -mg - \frac{v^2}{10}$$

$$a = \frac{dv}{dt} = -g - \frac{v^2}{10m} = -10 - \frac{v^2}{100} \quad (m = 10 \text{ kg})$$

$$10 + \frac{v^2}{100} = -dt$$

$\therefore$ New equation

$$2x\left(\frac{1}{3}\right)^2 - 9\left(x^{1/3}\right)^2 + 7 = 0$$

$$2x^{2/3} - 9x^{2/3} + 7 = 0$$

$$9x^{2/3} = 2x + 7$$

$$(9x^{2/3})^3 = (2x + 7)^3$$

$$729x^2 = 8x^3 + 84x^2 + 294x + 343$$

$$\Rightarrow 8x^3 - 645x^2 + 294x + 343 = 0$$

d) Solve $x^5 + 2x^4 - 2x^3 - 8x^2 - 7x - 2 = 0$

of multiplicity 4

$$P(x) = x^5 + 2x^4 - 2x^3 - 8x^2 - 7x - 2$$

$$P'(x) = 5x^4 + 8x^3 - 6x^2 - 16x - 7$$

$$P''(x) = 20x^3 + 24x^2 - 12x - 16$$

$$P'''(x) = 60x^2 + 48x - 12 = 0$$

$$12[5x^2 + 4x - 1] = 0$$

$$12[(5x-1)(x+1)] = 0$$

$$x = \frac{1}{5} \quad x = -1$$

$$a) \frac{x^2}{25} + \frac{y^2}{16} = 1$$

$$a^2 = 25 \quad b^2 = 16$$

$$a = 5 \quad b = 4$$

$$b^2 = a^2(1 - e^2)$$

$$16 = 25(1 - e^2)$$

$$16 = 25 - 25e^2$$

$$25e^2 = 25 - 16$$

$$e^2 = \frac{9}{25}$$

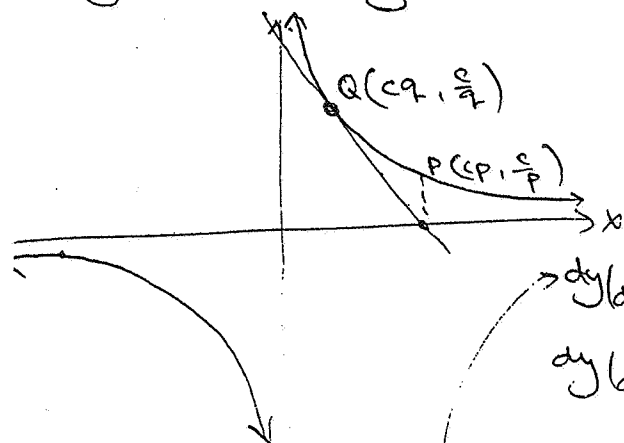
$$e = \frac{3}{5}$$

generally $S(ae, 0) \quad S'(-ae, 0)$

$$\therefore S(3, 0) \quad S'(-3, 0)$$

2

b) P and Q are variable points on rectangular hyperbola. $xy = c^2$



$$\frac{dy}{dx} = -c^2 x^{-2}$$

$$\frac{dy}{dx} = -\frac{c^2}{x^2}$$

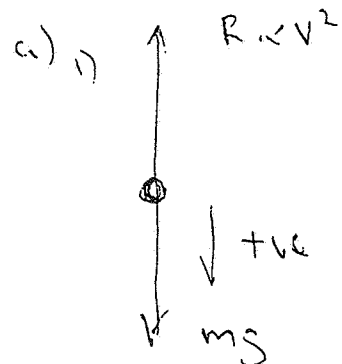
$$\text{at Q gradient} = \frac{-c^2}{c^2 q^2}$$

$$m = -\frac{1}{q^2}$$

$$xy = c^2$$

$$y = \frac{c^2}{x}$$

$$y = c^2 x^{-1}$$



$R = mkv^2$ let m be constant

$$1) \quad m\ddot{x} = mg - mkv^2$$

$$\ddot{x} = g - kv^2$$

$$\text{using } \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

$$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = g - kv^2$$

$$\frac{d}{dx} (v^2) = 2g - 2kv^2$$

1) Show that $v^2 = \frac{g}{k} - Ae^{-2kx}$ satisfies (1)

$$v^2 = \frac{g}{k} - Ae^{-2kx}$$

$$\frac{d}{dx} (v^2) = \frac{d}{dx} \left(\frac{g}{k} - Ae^{-2kx} \right)$$

Differentiate R.H.S.

$$\frac{d}{dx} (v^2) = 2Ake^{-2kx}$$

$$= 2kAe^{-2kx}$$

$$= 2k \left(\frac{g}{k} - v^2 \right)$$

$$= 2g - 2kv^2$$

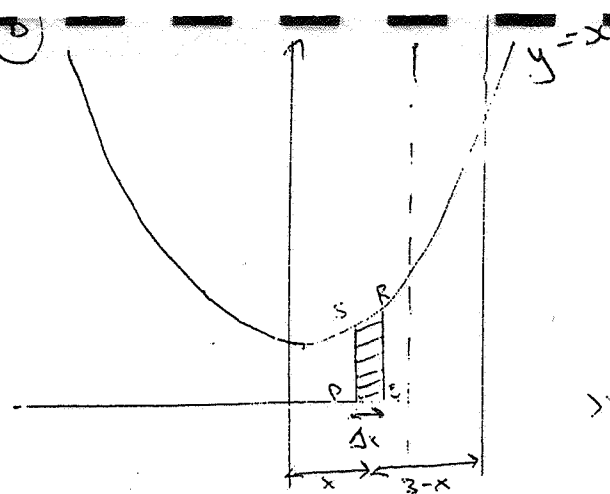
$$v^2 = \frac{g}{k} - Ae^{-2kx}$$

$$\therefore Ae^{-2kx} = \frac{g}{k} - v^2$$

when $x=0 \quad v=0$

$$\therefore 0 = \frac{g}{k} - Ae^0$$

$$\therefore A = \frac{g}{k}$$



$$V = \pi \int_0^3 [(3-x)^2 - (3-(x+\Delta x))^2] y$$

$$= \pi [6\Delta x - 2x\Delta x - \Delta x^2] y$$

$(\Delta x)^2$ negl

$$= \pi (6-2x) y \Delta x$$

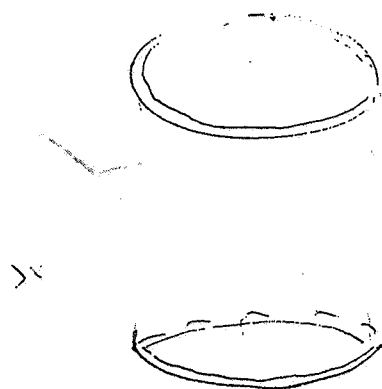
$$= \pi (6-2x)(x^2+6x+9) \Delta x$$

hence

$$V = \pi \int_0^3 (6x^2 - 2x^3 + 6x - x^2) dx$$

$$= \pi \left[2x^3 - \frac{x^4}{2} + 6x - x^2 \right]_0^3$$

$$= 16\pi$$



strip PQRS rotated about $x=3$

$$C = 2\pi r$$

inner radius = $3-(x+\Delta x)$

outer radius = $3-x$

$\therefore$ tangent at Q is

$$y - \frac{c}{q} = -\frac{1}{q^2} (x - cq)$$

$$x + q^2 y = 2cq$$

Cut x -axis at $y=0$

at this point $x=cq$ ie $P(cq, \frac{c}{q})$

$$\therefore cq = 2cq$$

$$\boxed{p = 2q}$$

3

ii) locus

Midpoint of PQ

$$x = \frac{cq + cq}{2}$$

$$y = \frac{\frac{c}{q} + \frac{c}{q}}{2}$$

$$x = \frac{c(p+q)}{2}$$

$$y = \frac{c}{2} \left(\frac{1}{p} + \frac{1}{q} \right)$$

Note for $p=2q$

$$x = \frac{3cq}{2}$$

$$2x = 3cq$$

$$q = \frac{2x}{3c}$$

$$y = \frac{c}{2} \left(\frac{1}{2q} + \frac{1}{q} \right)$$

$$y = \frac{c}{2} \left(\frac{1+2}{2q} \right)$$

$$y = \frac{c}{2} \left(\frac{3}{2q} \right)$$

$$y = \frac{3c}{4q}$$

$$4qy = 3c$$

$$q = \frac{3c}{4y}$$

$$\int \tan^{-1} x \, dx$$

$$a) i) \int \tan^{-1} x \, dx \rightarrow 2 \int \frac{dx}{2 \sin x \cos x}$$

$$\int 1 \cdot x \cdot \tan^{-1} x \, dx \Rightarrow 2 \int \frac{dx}{\sin 2x}$$

$$u = \tan^{-1} x \quad \frac{dv}{dx} = 1$$

$$\frac{du}{dx} = \frac{1}{1+x^2} \quad v = x$$

$$\Rightarrow 2 \int \frac{1}{\sin 2x} dx$$

$$\Rightarrow 2 \int \operatorname{Cosec} 2x \, dx$$

$$\Rightarrow \frac{2 \operatorname{Cosec} 2x (\operatorname{Cosec} 2x + \cot 2x)}{\operatorname{Cosec} 2x + \cot 2x}$$

$$\Rightarrow -\ln (\operatorname{Cosec} 2x + \cot 2x)$$

3
—

$$ii) \int \frac{dx}{\sin x \cos x}$$

$$\Rightarrow \int \frac{2}{2 \sin x \cos x} dx$$

$$\int -1 + \frac{8}{1+x^2} dx$$

Now consider, using partial fractions, $\frac{8}{4-x^2}$

$$\frac{8}{(2-x)(2+x)} = \frac{a}{2-x} + \frac{b}{2+x}$$

$$8 = a(2+x) + b(2-x)$$

$$\text{for } x = -2 \Rightarrow 4b = 8 \quad b = 2$$

$$\text{for } x = 2 \Rightarrow 4a = 8 \quad a = 2$$

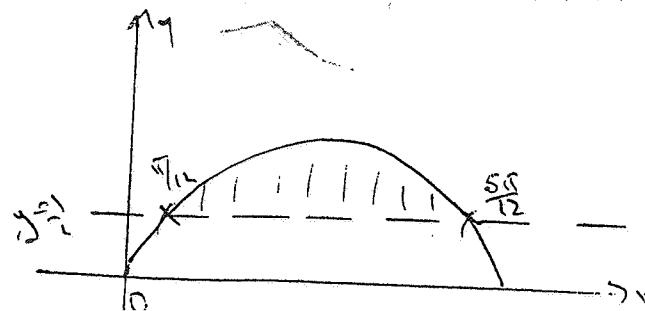
$$\Rightarrow -\int_{-1}^1 1 \, dx + \int_{-1}^1 \frac{2}{2-x} \, dx + \int_{-1}^1 \frac{2}{2+x} \, dx$$

$$\Rightarrow \left[-x - 2 \ln(2-x) + 2 \ln(2+x) \right]_{-1}^1$$

$$\Rightarrow -4 \ln 3 - 2$$

2c) Sketch region bounded by $y = \sin 2x$.

$$y = \frac{1}{2} \quad \text{region } 0 \leq x \leq \frac{\pi}{2}$$



Question Six

a) Determine real values for k in equation

$$\frac{x^2}{19-k} + \frac{y^2}{7-k} = 1$$

for an ellipse

$$19-k > 0 \text{ and } 7-k > 0$$

$$\therefore k < 19 \text{ and } k < 7$$

$$\therefore k < 7$$

for hyperbola

$$19-k > 0 \text{ and } 7-k < 0$$

$$k < 19 \text{ and } k > 7$$

$$\therefore 7 < k < 19$$

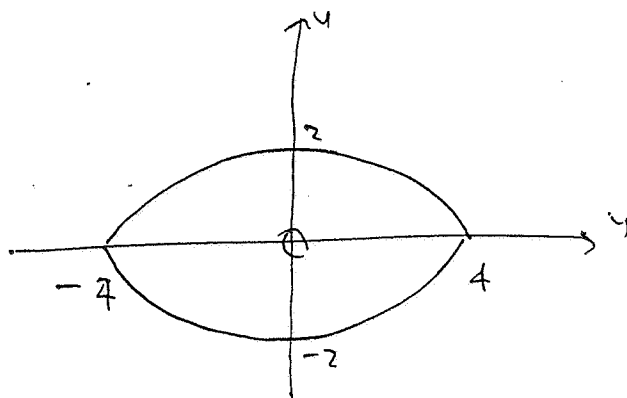
for $k=3$ equation becomes

$$\frac{x^2}{16} + \frac{y^2}{4} = 1$$

of the form

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$a=4 \quad b=2$$



b) Evaluate

$$\int_{-1}^1 \frac{4+x^2}{4-x^2} dx$$

by long division

$$\begin{array}{r} 1-x^2 \overline{) 4+x^2} \\ \underline{4-x^2} \\ 8x^2 \end{array}$$

$$= -1 + \frac{8}{4-x^2}$$

$$(ii) \int \frac{\sec^2 x \cdot dx}{\tan^2 x - 3 \tan x + 2}$$

$$\text{let } u = \tan x$$

$$\therefore \frac{du}{dx} = \sec^2 x$$

$$du = \sec^2 x \cdot dx$$

$$\Rightarrow \int \frac{du}{u^2 - 3u + 2}$$

$$\int \frac{1}{(u-2)(u-1)} du$$

Now consider partial fractions.

$$\frac{1}{(u-2)(u-1)} = \frac{a}{u-2} + \frac{b}{u-1}$$

$$1 \equiv a(u-1) + b(u-2)$$

$$\text{for } u=1 \quad b=-1$$

$$\text{for } u=2 \quad a=1$$

3
—

$$\Rightarrow \int \left(\frac{1}{u-2} - \frac{1}{u-1} \right) du$$

$$\ln(u-2) - \ln(u-1) + C$$

$$\text{but } u = \tan x$$

$$\therefore \ln(\tan x - 2) - \ln(\tan x - 1) + C$$

1) $\int_0^1 \frac{dx}{x^2 + 8x + 4}$

Complete the

$$\begin{aligned} x^2 + 8x &= -4 \\ x^2 + 8x + (4)^2 &= -4 + (4)^2 \\ (x+4)^2 &= 12 \\ (x+4)^2 - 12 &= 0 \end{aligned}$$

horizontal asymptote

$$\begin{aligned} y &= \frac{x^2}{x^2} - \frac{6x}{x^2} + \frac{13}{x^2} \\ &= \frac{x^2}{x^2} - \frac{6}{x} + \frac{13}{x^2} \end{aligned}$$

$$\Rightarrow \int_0^1 \frac{dx}{(x+4)^2 - 12}$$

$$\Rightarrow \frac{1}{2\sqrt{12}} \left[\ln \frac{x+4-\sqrt{12}}{x+4+\sqrt{12}} \right]_0^1$$

$$\Rightarrow \frac{1}{4\sqrt{3}} \left[\ln \frac{5-\sqrt{12}}{5+\sqrt{12}} - \ln \frac{4-\sqrt{12}}{4+\sqrt{12}} \right]$$

$$\Rightarrow \frac{1}{4\sqrt{3}} \left[\ln \left(\frac{5-\sqrt{12}}{5+\sqrt{12}} \times \frac{4+\sqrt{12}}{4-\sqrt{12}} \right) \right]$$

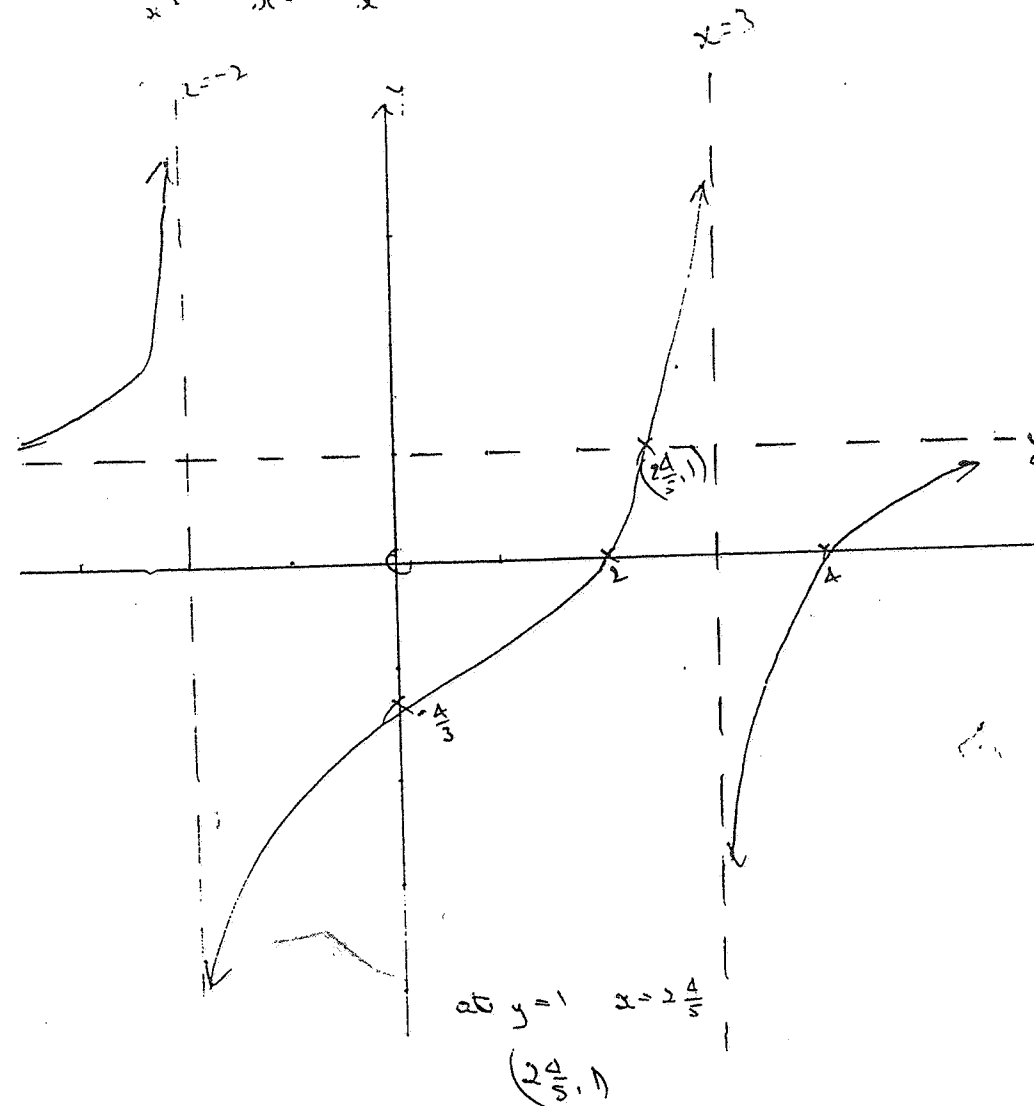
$$\Rightarrow \frac{1}{4\sqrt{3}} \ln \frac{20+2\sqrt{3}-12}{20-2\sqrt{3}-12}$$

$$\Rightarrow \frac{1}{4\sqrt{3}} \ln \frac{4+\sqrt{3}}{4-\sqrt{3}}$$

11) $\int_0^1 x^2 e^{-x} dx$

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

$$u = x^2 \quad \frac{dv}{dx} = e^{-x}$$



* Cuts x-axis at $y=0$

$$x^2 - 6x + 8 = 0$$

$$(x-2)(x-4) = 0$$

$$(2,0) (4,0)$$

Cut y-axis at $x=0$

$$\therefore y = -\frac{8}{6} = -\frac{4}{3}$$

$$\therefore (0, -\frac{4}{3})$$

* Stationary Values

$$y = \frac{u}{v}$$

$$u = x^2 - 6x + 8 \quad v = x^2 - x - 6$$

$$u' = 2x - 6 \quad v' = 2x - 1$$

$$\frac{dy}{dx} = \frac{(x^2 - x - 6)(2x - 6) - (x^2 - 6x + 8)(2x - 1)}{(x^2 - x - 6)^2}$$

$$\frac{dy}{dx} = \frac{5x^2 - 28x + 44}{(x^2 - x - 6)^2}$$

$$\frac{dy}{dx} = 0 \text{ when } 5x^2 - 28x + 44 = 0$$

Consider $b^2 - 4ac$

$$28^2 - 4 \times 5 \times 44$$

$$784 - 880$$

$$b^2 - 4ac < 0$$

* asymptotes
vertical

1) when $x^2 - x - 6 = 0$

$$(x+2)(x-3) = 0$$

$$x = -2 \quad x = 3$$

approaching $x=3$

for $x < 3$

$$y = \frac{(x-4)(x-2)}{(x+2)(x-3)}$$

$$y = \frac{-+}{++} = +$$

y is large

$$x < 3 \quad y = \frac{+-}{++} = -$$

y large -ve.

$x > 3$

$$y = \frac{--}{++} = +$$

y large -ve.

$$= [-x^2 \cdot e^{-x}]_0^1 + \int 2x \cdot e^{-x} dx$$

2nd application.

$$u = 2x \quad \frac{dv}{dx} = e^{-x}$$

$$\frac{du}{dx} = 2$$

$$v = -e^{-x}$$

$$= -2x e^{-x} + \int_0^1 2 e^{-x} dx$$

$$= -2x \cdot e^{-x} - 2(e^{-x})_0^1$$

$$\Rightarrow [-x^2 \cdot e^{-x}]_0^1 - [2x \cdot e^{-x}]_0^1 - 2[e^{-x}]_0^1$$

$$\Rightarrow -\frac{1}{e} - 2e^{-1} - 2e^{-1} + 2$$

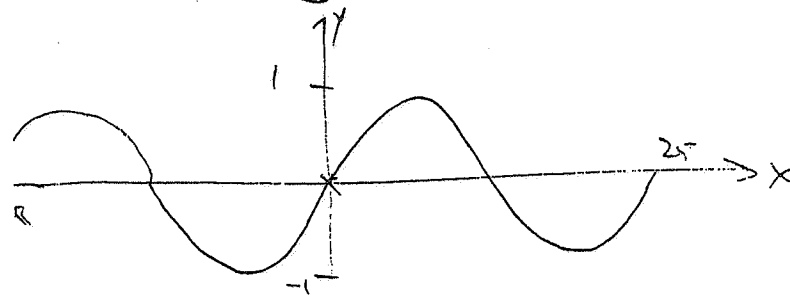
$$= 2 - 5e^{-1}$$

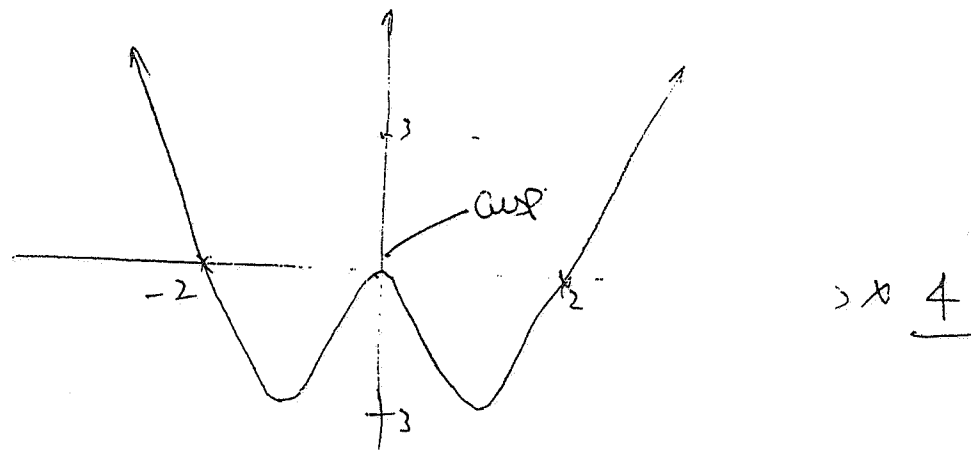
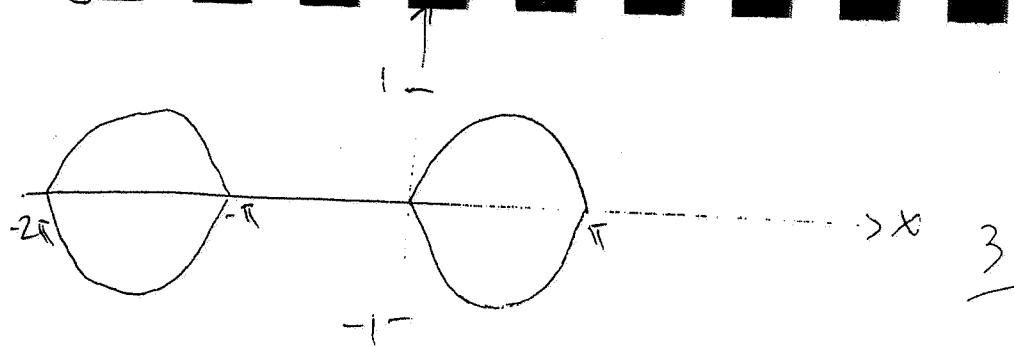
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Question 5

$$-2\pi \leq x \leq 2\pi$$

a) Sketch $y = \sin x$ and hence $y^2 = \sin$





b) Sketch $y = x^3 - 4x$.

Cut y-axis at $x=0 \therefore y=0$

Cut x-axis at $y=0$

$$0 = x(x^2 - 4)$$

$$x=0 \quad x=\pm 2$$

Stationary Values

$$dy/dx = 3x^2 - 4 = 0$$

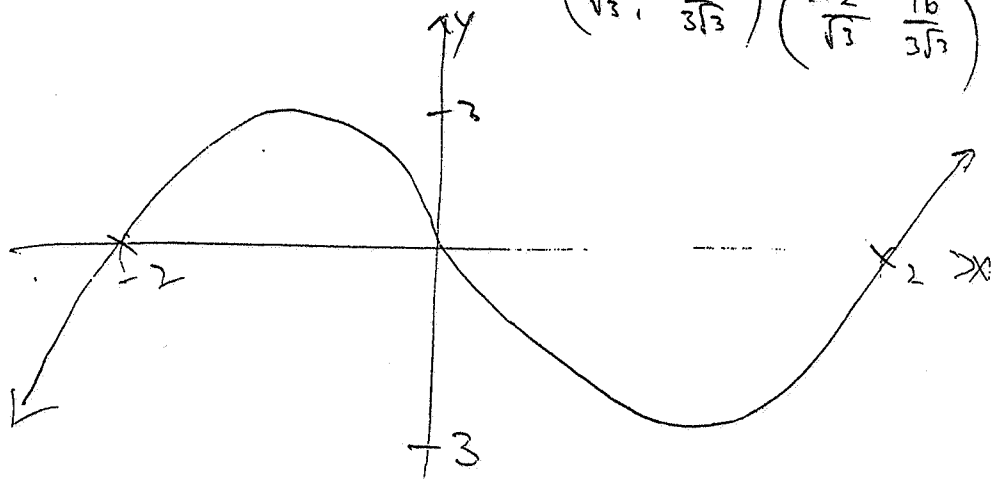
$$3x^2 = 4$$

$$x^2 = \frac{4}{3}$$

$$x = \pm \frac{2}{\sqrt{3}}$$

$$\therefore y = \pm \frac{16}{3\sqrt{3}}$$

$$\therefore \text{S.V.} \left(\frac{2}{\sqrt{3}}, -\frac{16}{3\sqrt{3}} \right) \left(-\frac{2}{\sqrt{3}}, \frac{16}{3\sqrt{3}} \right)$$



or by subtraction of ordinates.

$$y_1 = |x|^3 \quad y_2 = 4|x|$$

$$\therefore y_1 - y_2$$

x	-3	-2	-1	0	1	2	3
y ₁	27	8	1	0	1	8	27
y ₂	12	8	4	0	4	8	12
y ₁ - y ₂	15	0	-3	0	-3	0	15

Sketch as shown above.

c) Sketch $y = \frac{x^2 - 6x + 8}{x^2 - x - 6}$